RATIO ANALYSIS

- Profitability ratios
- Return ratios
- Activity ratios (Efficiency)
- Leverage ratios (Solvency)
- Liquidity ratios
- Du Pont analysis (RoE breakup)
- For each ratio, think about:
  - What aspect(s) of business is the ratio measuring?
  - High value is good news or low value is good news?

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PROFITABILITY RATIOS

- Gross Profit Margin = Gross Profit / Sales
- EBITDA Margin = EBITDA / Sales
- EBIT Margin (Operating Profit margin) = EBIT / Sales
- NOPAT Margin = EBIT(1-t) / Sales
  - NOPAT = Net Operating Profit After Taxes = EBIT(1-t)
- Profit Margin (Return on Sales) = Net Profit / Sales
- "Sales" is usually "Revenue from Operations"

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RETURN RATIOS

- Return on Assets (RoA) = EBIT (1-t) / Avg Total Assets
  - Textbook has PAT / Avg Total Assets
- Return on Equity (RoE) = Net Profit / Avg Networth
  - Networth = Shareholders Equity = Equity
- Return on Cap Employed (RoCE) = EBIT(1-t) / Avg CE
  - CE = Networth + Debt – C&CE (non Operating Assets)
  - "Other" income excluded in numerator
  - Pronounced roki or rozy. (Is the company's RoCE rosy? ☺)
  - Sometimes, RoCE is defined on pre-tax basis

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ACTIVITY RATIOS (EFFICIENCY)

- Asset Turnover = Sales / Avg Total Assets
- Fixed Asset Turnover = Sales / Avg Fixed Assets
- Receivables Turnover = Sales / Avg Trade Receivables
  - Avg Collection Period (ACP) = Avg TR / Daily Cr Sales
  - ACP = 365/Receivables Turnover
- Inventory Turnover = CoGS / Avg Inventory
  - Inv Holding Period (IHP) = Avg Inventory / Daily CoGS
  - IHP = 365/Inventory Turnover
- Payables Turnover = Purchases / Avg Trade Payables
  - Avg Payment Period (APP) = Avg TP / Daily Cr Purchases
  - CoGS is used in case data on Purchases is not available
  - APP = 365/Payables Turnover
- Operating Cycle = IHP + ACP
- Net Operating Cycle = IHP + ACP – APP
  - Also called Cash Cycle or Cash Conversion Cycle

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LEVERAGE RATIOS (SOLVENCY)

- Debt-Equity ratio = Debt / Networth
  - “Debt” generally means long term interest-bearing borrowings
- Interest Coverage Ratio = EBIT / Interest expense (Finance cost)

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LIQUIDITY RATIOS

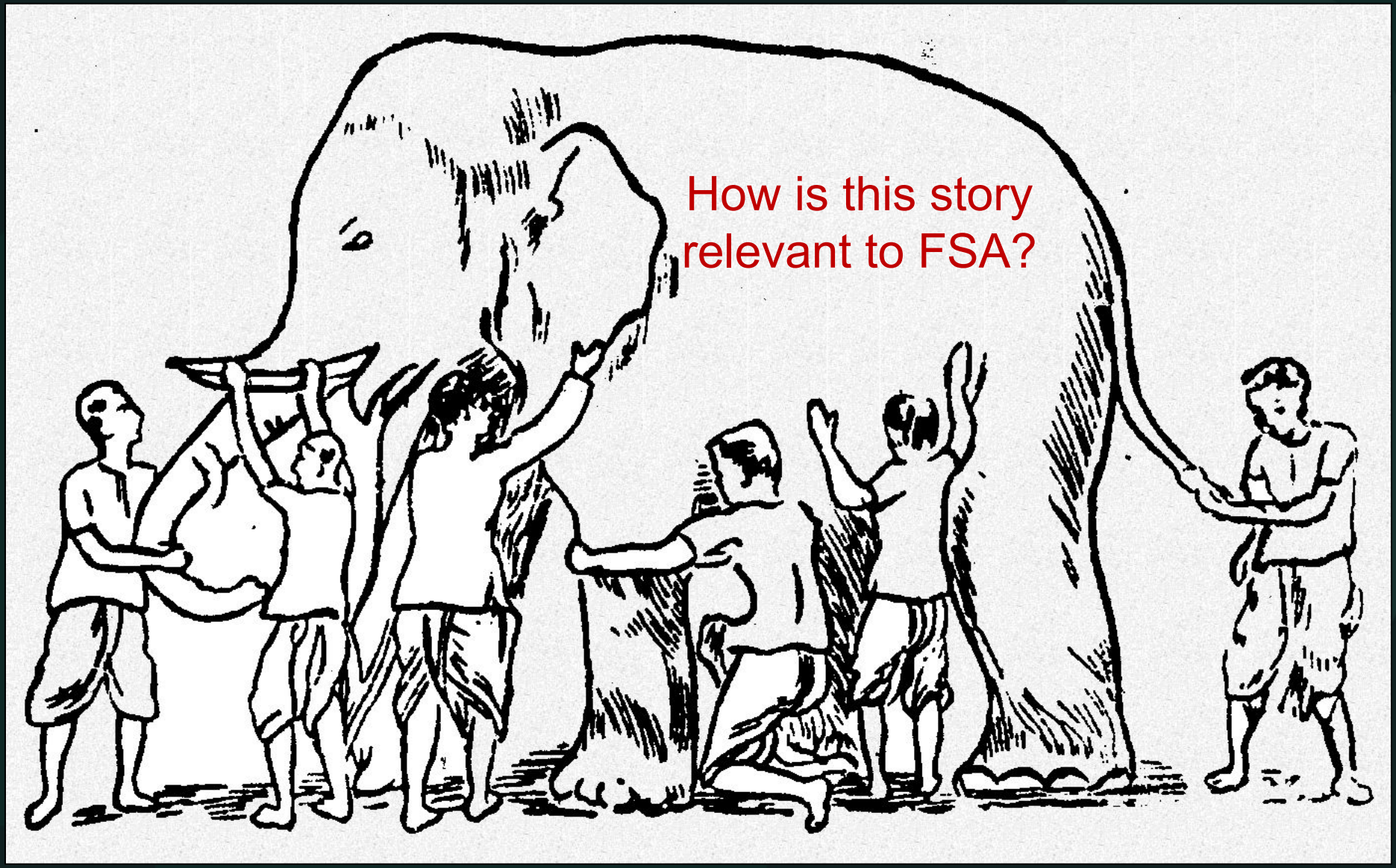
- Current ratio = Current Assets / Current Liabilities
- Quick ratio = (Cur Assets – Inventory) / Cur Liabilities
- Net Working Capital (NWC or WC) = CA - CL

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DU PONT ANALYSIS

- Developed in the DuPont Corp in 1910s
- RoE = Net Profit / Networth
- $\text{RoE} = \frac{Net\ Profit}{Sales} * \frac{Sales}{Total\ Assets} * \frac{Total\ Assets}{Networth}$
- Profit Margin * Asset Turnover * Equity Multiplier
- Profitability * Efficiency * Leverage
- What is driving RoE?
- RoE = RoA * Leverage
  - RoA is defined as PAT/Assets for this reason
- 3 levers to influence RoE
- Excel examples (Hypothetical and Adani Green)

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https://jainworld.com/education/jain-education-material/jain-stories/elephant-and-the-blind-men/

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PROBLEMS

Review Problem

In 2016, the following parties have to act based on their evaluation of DRL's financial statements:

- Maharashtra Bank is examining a request from DRL for long-term financing.
- Bharat Chemical Company is considering the supply of raw materials on credit to DRL.
- Arvind Shah is thinking whether to invest his retirement benefits in DRL's shares.

Industry averages for illustrative purposes for selected ratios are given below:

Profit margin	4.2%
Asset turnover	1.7 times
Return on assets	7.4%
Return on equity	21%
Earnings per share	`4.21
Current ratio	1.46
Quick ratio	1.12
Debt-to-equity	2.3
Interest cover	4.13
Price-earnings ratio	27 times

For DRL	2016	2015
PM	13.71%	15.55%
ATO (times)	0.81	0.87
RoA	11.14%	13.49%
RoE	19.96%	26.37%
EPS (Rs.)	126.15	137.18
Cur ratio	1.73	1.74
Quick ratio	1.35	1.36
D/E(times)	0.29	0.37
Int cover	39.07	27.80
MPS (Rs.)	3,085	3,438

Required

1. Select two or three indicators that would be most relevant to the three decision makers.
2. Based on your analysis of the indicators, what would you recommend?

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